Lugar das raízes

$$T = \frac{KG}{1+KGH}$$

Malha fechada

$$1+KGH=0$$

$$KGH = -1 = 1 \angle 180^\circ (2k+1)$$

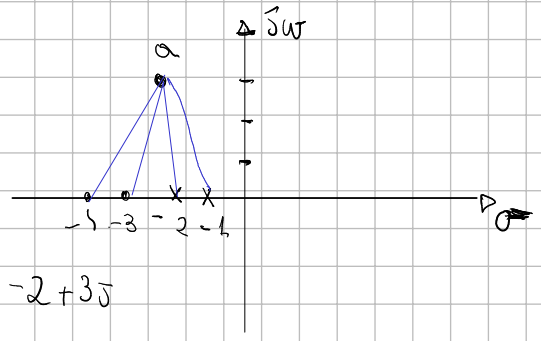
$k = 0, \pm 1, \pm 2 \ldots$

$$K = \frac{1}{|G||H|}\Bigg|_{s=a}$$

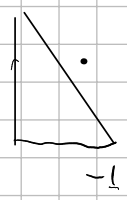
$$KGH = \frac{K(s+3)(s+4)}{(s+1)(s+2)}$$

zeros m. aberta: $-3$ e $-4$

polos m aberta: $-1$ e $-2$




$a = -2+3j$




$$\text{arctg}\left(\frac{\text{imag}(a-(-1))}{\text{real}(a-(-1))}\right)$$

Contribuição angular

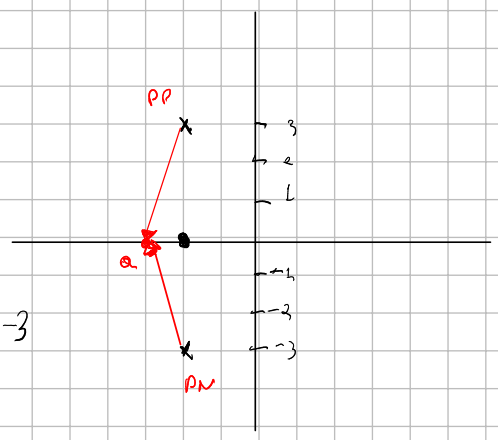
$$\sum \theta_z - \sum \theta_p$$

# Outro Exercício

$$KGH = \frac{K(s+2)}{s^2+4s+13}$$

Zeros M Aberta

Polos M. Aberta $= -2 \pm j3$




$Q = -3$

$\angle Z_Q = 0 \Rightarrow \angle Z_{QC} = 180$

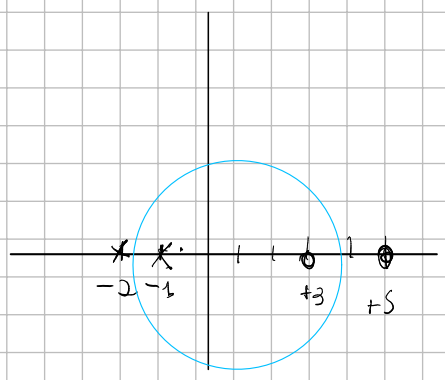
$\angle PP$ $\qquad$ $\angle PP_C\ (3^\circ Q) =$

$\angle PN$ $\qquad$ $\angle PN_C\ (2^\circ Q) =$

$$\sum \frac{1}{\sigma + z_i} = \sum \frac{1}{\sigma + P_i}$$

$$\frac{1}{\sigma-3} + \frac{1}{\sigma-5} = \frac{1}{\sigma+2} + \frac{1}{\sigma+1}$$

$$\frac{1}{\sigma-3} + \frac{1}{\sigma-5} - \frac{1}{\sigma+2} - \frac{1}{\sigma+1} = 0 = Y$$



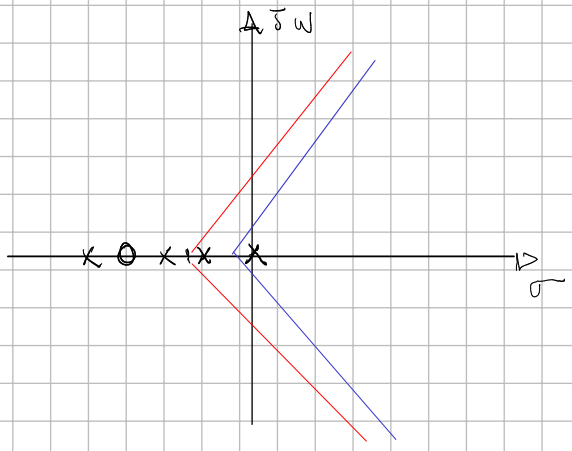

$$\sigma = \begin{cases} +3{,}8 & \checkmark \\ -1{,}445 & \checkmark \end{cases}$$

# Exercício

$$KG = \frac{K(s+3)}{s(s+1)(s+2)(s+4)}$$

Zero M.A $= -3$

Polo M.A $= 0, -1, -2, -4$



$$T(s) = \frac{\qquad}{s^4}$$

4 RAMOS

$$\frac{1}{\sigma+3} = \frac{1}{\sigma} + \frac{1}{\sigma+1} + \frac{1}{\sigma+2} + \frac{1}{\sigma+4}$$

$$\sigma \begin{cases} -3,3 \pm 0,685\,j \\ -1,61 \quad X \\ -0,43 \quad V \end{cases}$$

Assintotas

$$\sigma_A = \frac{0-1-2-4-(-3)}{4-1} = \frac{4}{3} = 1,33$$

$$\theta_A = \frac{(2k+1)\cdot 180}{4-1}$$

$k=0 \Rightarrow \theta_A = 60°$

$k=1 \Rightarrow \theta_A = 180°$

$k=2 \Rightarrow \theta_A = 300°$ ou $-60°$